Task # 06:

Scenario: Library Management System

Sample Tables Data

1. **Books Table**

BookId	Title	Author	Category	Price	Quantity
1	Harry Potter	J.K. Rowling	Fiction	899.99	5
2	Data Science 101	John Smith	Education	49.99	12
3	The Great Gatsby	F. Scott	Classic	599.99	2

2. **Users Table**

UserId	Name	Email	City	Membership_status
101	Alice Brown	broalice73@gmail.com	New York	Active
102	Bob Green	bobgreen@nu.edu.pk	Chicago	Active
103	Charlie Lee	leeee72@yahoo.com	Null	Inactive

3. **Transactions Table**

TransactionId	book_id	borrower_id	borrow_date	return_date	status
1	1	101	2025-01-10	2024-02-20	Returned
2	3	102	2024-12-22	NULL	Overdue
3	3	101	2025-02-15	NULL	Borrowed

Insert at least 10 records in each table; books, users and transactions.

Run following Queries:

1. Choose one book and update its category to Poetry.
2. List all the user ids and names for the users who haven't provided their cities.
3. Considering due date of 1 week, update status to overdue for overdue books.
4. Display all book ids in descending order that are not returned yet.
5. List all book where the price is more than $250 and quantity is below 5.
6. Display the transactionId and status of 3 most recent transactions.
7. Retrieve distinct categories of books from the `Books` table.
8. Update the price of all books in the 'Education' category to decrease by 10%.
9. Select books where the price is either less than $50 or quantity is between 5 and 10.
10. List all book titles that do not have 'w' as the third letter and are currently out of stock
11. Find all books that have 'data' in their name and do not belong to the Fiction Category.
12. Find all users who belong from New York, Chicago or Seattle as 'CustomerCity' without using OR operator.
13. Find all customers whose email addresses do not follow a valid format.
14. Create a new table ArchivedMembers with attributes ArchivedUserId, name, email and city. Then transfer all data for the inactive members from the users table.
15. Delete all inactive members from the database.